

Nicolás Saldarriaga Carvajal

FULL-STACK DEVELOPER · FINANCIAL DATA · PYTHON
/ FLASK / POSTGRESQL / AZURE

I design, build and run the software that finance teams decide with. In 2026 I built and now operate two production web platforms for Prisier, working alone. Before that, I built data tools for the IDB Treasury and for IFAD's Latin America portfolio.

I know how treasury, portfolio and risk teams use their data, and I build the systems that serve it.

11 days

To design, test and deploy Prisier One on Microsoft Azure (BOps live on Azure since Apr 2026)

409

Commits · Jul–Oct 2026

~69k

Lines of Python in production

~740

Automated tests

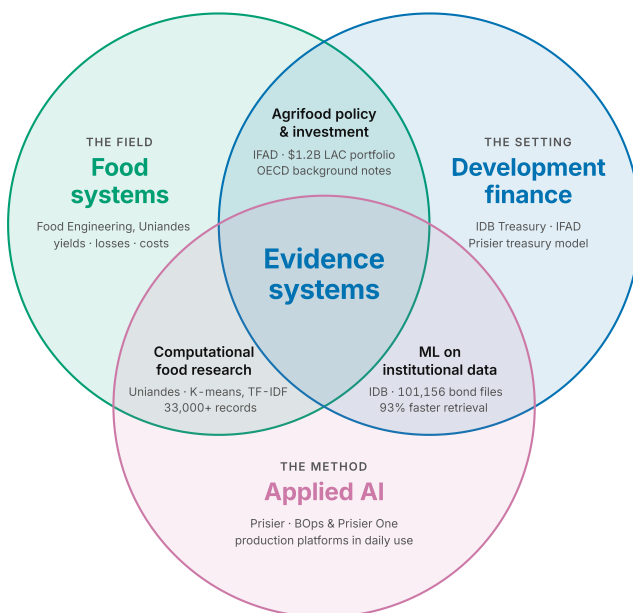
101,156

IDB investment files classified with ML

\$1.2B+

IFAD portfolio I automated tracking for

1) Profile · Where finance, data and software overlap



Evidence architect for food & development finance

Evidence that moves policy and capital.

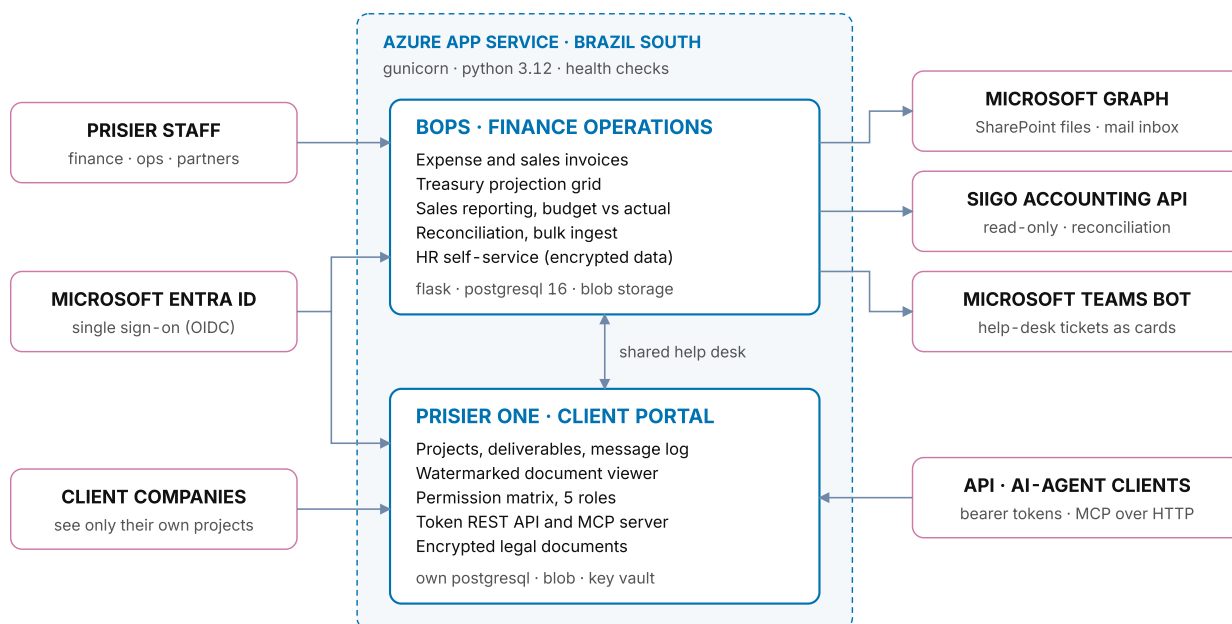
I turn the messy information institutions sit on into evidence they can decide with. I learned the questions inside IFAD, the IDB and the OECD, and I build the systems, now **with AI**, that answer them.

- **Food systems** is the field
- **Development finance** is the setting
- **Applied AI** is the method
- **Evidence systems** is what I deliver

2) Prasier · Two production platforms on Azure Feb 2026 – present · Bogotá

Prasier is a pricing and revenue-management consultancy. I designed, built, deployed and maintain both of its internal platforms. BOps runs the firm's finances; Prasier One is where clients receive their deliverables.

Architecture 2 apps · isolated data · 1 sign-in · 1 help desk



BOps · back-office finance

241 commits · ~42k lines · 56 models

- **Invoices:** expense and sales invoices from upload, bulk zip import or a monitored mailbox, with duplicate checks, extraction and an approval step.
- **Treasury:** a weekly, monthly and yearly cash projection. Every cell is computed from invoice and debt data, and the balance is anchored to real bank balances.
- **Financial model:** generates an Excel workbook with live formulas and three cash scenarios (committed, weighted, full pipeline) from a read-only database transaction.
- **Reconciliation:** a one-way sync with the accounting system. It changes a record only when the match is unambiguous, and every change is audited.
- **Dashboards:** sales by client and month, budget vs actual, pipeline. Manual overrides are flagged when they differ from the computed value.

Prasier One · client portal

168 commits · ~28k lines · ~740 tests

- **Access control:** one permission matrix is the single source of truth, and a test restates it independently. The app refuses to start if a client could ever reach internal tools.
- **Secure viewer:** documents are served in a sandboxed frame with a watermark naming the viewer and the time. Permission is rechecked on every load.
- **API and MCP:** a token REST API and an MCP server let staff publish deliverables from scripts and AI agents.
- **Operations:** scheduled maintenance windows, release notices, bulk account creation and audited deletes behind three confirmations.

Source code is private and belongs to Prasier SAS. A live walkthrough of both platforms is available on request.

3) Results · What changed, measured Measured on real company data

Metric	Before → after	What I built and how it was measured
Wrong invoice totals	19% → 5.3%	Replaced a vision - AI reader with a deterministic PDF parser that accepts a value only when two independent methods agree. Held-out test: 281 of 778 confirmed invoices. Sales invoices now read 90% exact.
Help-desk wrong answers	18% → 7%	Rebuilt the help - desk bot as a search engine over the knowledge base that escalates to a person when unsure. Correct answers went from 54 to 86, with 0 of 25 false positives.
Invoice batch time	-30 min	Saved per 250 - invoice batch (about 7 s per invoice) by removing the AI model's rate - limit pause.
Cash-timing accuracy	227 / 395	Expense rows moved to their correct payment week in the treasury projection, with the total unchanged.
Document load time	13.2 s → 0.67 s	A 100 - page deck in the client portal.

4) Security hardening · built for financial and personal data OWASP ASVS Level 2 review

Identity and access

- Microsoft Entra single sign-on for staff. Users are matched on an immutable ID and accounts are never created automatically.
- Roles live in the app's own database: one permission matrix, checked on every request and restated in tests.
- Separate rights for exporting data, scoped per partner.

Data protection

- Personal and payroll data encrypted at the column level with a dedicated key. Without the key the module refuses to serve, so it never falls back to plain text.
- Legal documents stored with envelope encryption (AES-GCM).
- Client data kept on separate database, storage and key-vault resources. Both apps hosted in Brazil South for Latin American data residency.

Application and integrations

- Strict content security policy with zero inline scripts; I moved 128 inline handlers to remove the last exception.
- API tokens expire, can be scoped to one project and are stored only as hashes. The API is off by default and every action is audited.
- Least privilege on integrations: mailbox access is limited to one mailbox (verified: allowed there, refused elsewhere), the accounting API is read-only, and the help desk queries the database in read-only transactions.

Control

In place

Pre-deploy audit	Every route checked for an access guard before each release; must report zero unguarded routes.
Abuse protection	Rate limiting on sign-in and password reset, account lockout, CSRF protection, idle timeouts, forced sign-out when a password changes.
Uploads	File type checked from content rather than the extension, zip limits against decompression bombs, image-only attachments with SVG refused.
Audit trail	An append-only security log, an admin audit viewer and a compliance view of every account's permissions.
Resilience	Health checks, auto-heal, metric alerts, 35-day geo-redundant database backups, storage versioning, feature flags and kill switches per module.

5) Analytics · Pricing for a national apparel retailer Prisier client · Mar 2026

For a Prisier client with more than 230 stores in four countries, I analysed a full year of sales: **6.2M** transaction lines in a **1.3 GB** file, in Python (pandas, NumPy, matplotlib).

- Normalised mixed currencies and stores that record order totals instead of per-unit prices.
- Computed margin per transaction and broke it down by country, store, channel, product line and month.

The client's name and figures are confidential and left out on purpose.

- Found that **2.1%** of transactions were sold below cost, and traced the losses to a few stores where 86–98% of sales carried a deep discount.
- Delivered 10 charts, written reports and a presentation.

6) Inside the IDB, IFAD and the OECD

Inter-American Development Bank · Treasury

Washington D.C. · Jun–Aug 2025

- Classified **101,156** fixed-income investment files with machine learning, cutting file retrieval time by **93%**.
- Built a central data platform with **14,987** bond definitions and **13,846** strategy records (Python, VBA/Excel). Manual processing fell **60%**, and portfolio reporting became a repeatable process.

Python

ML classification

VBA

Fixed income

IFAD · Latin America and Caribbean

Rome · Jul 2024–Jan 2025

- Automated portfolio information flows and impact tracking for **45+** projects worth **\$1.2B+**. Request processing time fell **85%**, and the work informed **\$200M+** in new approvals.
- The same tools became a covenant-tracking system with **35+** KPIs per project, adopted across the region.
- Set quality-control protocols for 620+ knowledge products. The department adopted them.

Automation

KPI monitoring

Portfolio data

OECD · Trade and Agriculture Directorate

Bogotá · Feb–Dec 2026

- Co-organised two national workshops between Colombia's Ministry of Agriculture and the OECD in Bogotá (July 2026), with about **90** participants from ministries, statistical agencies, the private sector, FAO, the IDB and ECLAC. Also supported four expert focus groups across six countries.
- Wrote two background notes on Colombia, about **15,000** words each, both accepted and feeding OECD case studies: one on measuring food loss and waste, one on equal opportunities in food systems.
- Mapped Colombia's family farm unit for every municipality in Python, as a geospatial input to the policy analysis.

Stakeholder engagement

Policy analysis

Geospatial

Universidad de los Andes

Bogotá · Aug 2023–Jun 2024

- Mixed-methods impact study of **1,774** community initiatives and **33,000+** development records in post-conflict territories.
- Used K-means clustering and TF-IDF to find investment gaps. Submitted to *World Food Policy*.

K-means

TF-IDF

Mixed methods

7) Role fit · Requirements and evidence SRM / RMG · Developer (PEC)

The role asks for	Where I have done it
Full-stack web apps	BOPs and Priser One: designed, built, deployed and maintained alone, both in daily production use.
Interactive dashboards	Treasury projection grid, sales and budget dashboards, and a shared chart engine checked for accessibility. Power BI before that.
APIs	Token REST API and MCP server (Priser One); clients for the Siigo accounting API, Microsoft Graph and Teams (BOPs).
Databases	PostgreSQL schemas (56 models in BOPs), idempotent migrations, read-only reporting transactions; the IDB bond and strategy data platform.
Version control	Git and GitHub for all work: 409 commits across both platforms in under three months.
Code reviews	I review every change before it ships, with test suites and a pre-deploy route audit. I'm ready to join a team review process.
Risk and treasury domain	IDB Treasury fixed-income data, IFAD covenant monitoring, and Priser's treasury, collections and reconciliation modules.

8) Stack

Backend	Python 3.12, Flask, SQLAlchemy, Gunicorn, REST, MCP
Frontend	Jinja templates, JavaScript, HTML/CSS, a custom chart engine and design system
Data	PostgreSQL, SQL, pandas, NumPy, openpyxl, Excel/VBA, Power BI
Cloud	Azure App Service, Azure Database for PostgreSQL, Blob Storage, Key Vault, Entra ID, Bot Service, Microsoft Graph
Methods	ML classification, K-means, TF-IDF, BM25 retrieval, geospatial analysis, LLM applications, AI-assisted engineering
Languages	Spanish (native), English (fluent), French (conversational)

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Bogotá, Colombia · October 2026 · Figures as of October 2026